GAD vs Sella on HIP — Comprehensive IRC Validation

2026-04-17 — Narval A100 MIG, Transition1x train split, 300 samples × 6 noise levels

TL;DR. GAD and Sella, applied as TS-finding optimizers to HIP, land at TSs that an IRC integrator can reliably validate. When comparing their *full 300-sample* output, **GAD degrades gracefully with noise while Sella drops sharply**. At 10 pm both methods succeed on ∼92% of samples. At 200 pm, IRC-confirmed GAD is 47.9% of samples; IRC-confirmed Sella is 24.0%. Per-converged-TS IRC quality is essentially identical (∼92% TOPO either way) — the robustness gap is entirely in how many samples each optimizer successfully converges on at high noise.

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1 Methodology

1.1 Dataset and noise

Transition1x train split, samples 0–299 (a fixed 300-sample universe). Each sample is a labeled reactant / transition state / product triple computed at the ωB97x/6-31G(d) level.

Starting geometry. For each sample i and noise level $\sigma \in \{10, 30, 50, 100, 150, 200\}$ pm, we perturb the labeled TS coordinates with Gaussian noise:

$$\mathbf{x}_0^{(i,\sigma)} = \mathbf{x}_{\mathrm{TS}}^{(i)} + \sigma \cdot \boldsymbol{\xi}_i, \qquad \boldsymbol{\xi}_i \sim \mathcal{N}(0, I_{3N}). \tag{1}$$

Noise seed `torch.manual_seed(42)` then sequential `torch.randn_like`; this generates a deterministic $\{\boldsymbol{\xi}_i\}_{i=0}^{299}$ set shared across GAD and Sella runs.

Calculator. All forces, energies, and Hessians from HIP neural network potential (`hip_v2.ckpt`). Hessians are analytic. On A100 MIG (2g.10gb) slices, one Hessian evaluation takes ∼20 ms.

1.2 GAD (gad_dt003)

Projected GAD with fixed Euler timestep. Per step:

$$\mathbf{H}_{\mathrm{proj}} = P_{\mathrm{Eckart}}\, M^{-1/2}\, \mathbf{H}\, M^{-1/2}\, P_{\mathrm{Eckart}} \tag{2}$$

$$\mathbf{v}_1 = \text{lowest eigenvector of } \mathbf{H}_{\mathrm{proj}} \tag{3}$$

$$\mathbf{F}_{\mathrm{GAD}} = \mathbf{F} + 2\,(\mathbf{F} \cdot \mathbf{v}_1)\,\mathbf{v}_1 \tag{4}$$

$$\mathbf{x}_{n+1} = \mathbf{x}_n + \mathrm{d}t \cdot \mathbf{F}_{\mathrm{GAD}}, \quad \mathrm{d}t = 0.003 \tag{5}$$

Max 2000 steps per sample (200 pm sweep was partial — 259/300 due to SLURM timeout). All operations pure torch; autograd flows through projection.

GAD convergence criterion. Checked on the Eckart-projected vibrational Hessian:

$$\text{GAD-converged} \;\equiv\; \big(n_{\mathrm{neg}} = 1\big) \;\wedge\; \big(\|\mathbf{F}\|_{\mathrm{mean}} < 0.01 \text{ eV/Å}\big) \tag{6}$$

where $\|\mathbf{F}\|_{\mathrm{mean}} = \frac{1}{N}\sum_i |\mathbf{f}_i|$ (mean per-atom force magnitude).

1.3 Sella (cartesian + Eckart)

Sella's TS optimizer (quasi-Newton in a modified basis) in *Cartesian* coordinates, with HIP's analytic Hessian fed in every step and Eckart-projected. `delta0`=0.048, `diag_every_n`=1, `gamma`=0, 1000 max steps.

Sella convergence criterion. Same n_{neg} check, different force metric:

$$\text{Sella-converged} \;\equiv\; \big(n_{\mathrm{neg}} = 1\big) \;\wedge\; \big(f_{\max} < 0.01 \text{ eV/Å}\big), \quad f_{\max} = \max_i |\mathbf{f}_i| \tag{7}$$

Note $f_{\max} \geq \|\mathbf{F}\|_{\mathrm{mean}}$ always; for N-atom systems the gap is typically 2–4×.

1.4 IRC (sella_hip)

Sella's IRC integrator, Cartesian coords, dx=0.1 Å arc-length, η=10^{-4}, γ=0.4. At every inner kick, Sella's BFGS-updated Hessian is overwritten with HIP's analytic Hessian (mass-weighted → Eckart-projected → un-mass-weighted), and the IRC's own halt-at-step-0 behaviour is bypassed by forcing at least one `step()` call (the initial kick off the saddle). Up to 500 inner steps per direction.

1.5 IRC validation criteria

For a given IRC run we have the forward and reverse endpoint coordinates $\mathbf{x}_f, \mathbf{x}_r$, the labeled $\mathbf{x}_R, \mathbf{x}_P$, and the per-endpoint Eckart-projected vibrational spectrum.

- **Bond-graph topology (primary)**:

$$\begin{aligned}\texttt{TOPO-intended} &\equiv (g_f \cong g_R \wedge g_r \cong g_P) \vee (g_f \cong g_P \wedge g_r \cong g_R)\\ \texttt{TOPO-half} &\equiv (g_f \cong g_R \vee g_f \cong g_P \vee g_r \cong g_R \vee g_r \cong g_P) \wedge \neg\texttt{TOPO-intended}\end{aligned}$$

  where $g(\cdot)$ is the element-labeled bond graph (covalent-radius heuristic); $\cong$ is labeled graph isomorphism.
- **Strict RMSD** (< 0.3 Å, Kabsch+Hungarian):

$$\begin{aligned}d_{\to R} &= \min(d^{f\to R}, d^{r\to R}), \quad d_{\to P} = \min(d^{f\to P}, d^{r\to P})\\ \texttt{RMSD-intended} &\equiv (d_{\to R} < 0.3) \wedge (d_{\to P} < 0.3)\end{aligned}$$

- **Endpoint vibrational quality**:

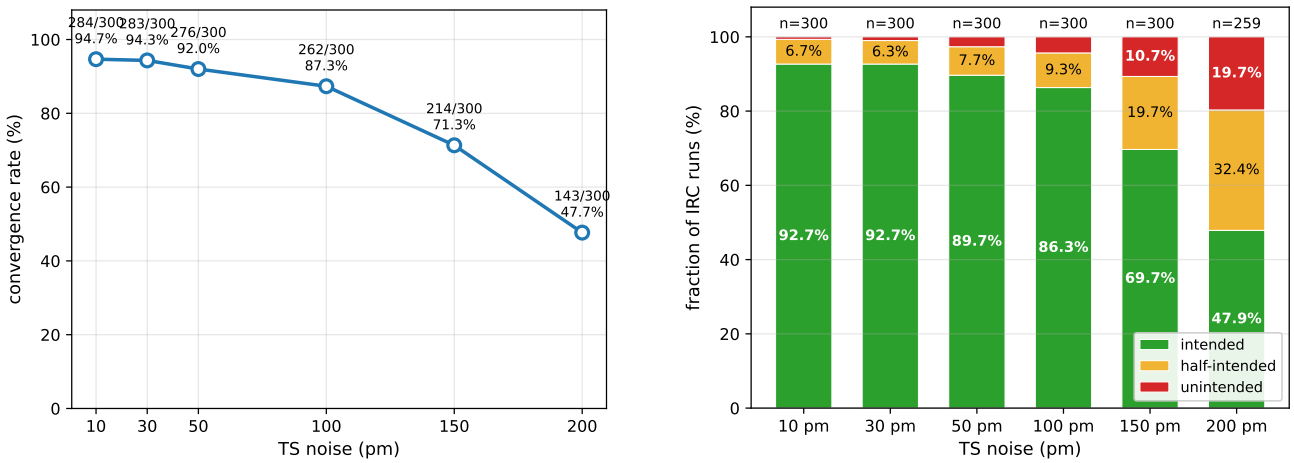
$$\begin{aligned}\texttt{both_at_min} &\equiv n_{\text{neg,vib}}(\mathbf{x}_f){=}0 \wedge n_{\text{neg,vib}}(\mathbf{x}_r){=}0\\ \texttt{neither_at_min} &\equiv n_{\text{neg,vib}}(\mathbf{x}_f){>}0 \wedge n_{\text{neg,vib}}(\mathbf{x}_r){>}0\end{aligned}$$

  Measures only whether IRC descended to a real vibrational minimum — does not reference labels.

2 Main Results

The core question: when we ask GAD or Sella for a TS on sample i at noise σ, how often does the method's own convergence criterion fire, and how often does an independent IRC check (on the final geometry) confirm the TS is chemically correct?

GAD



(a) GAD convergence rate vs. TS noise. y = fraction of the 300-sample universe where `gad_dt003` reached $n_{\text{neg}}=1 \wedge \|\mathbf{F}\|_{\text{mean}} < 0.01$ (Eq. 6) within 2000 steps at dt=0.003. Annotations: converged / 300, converged%.

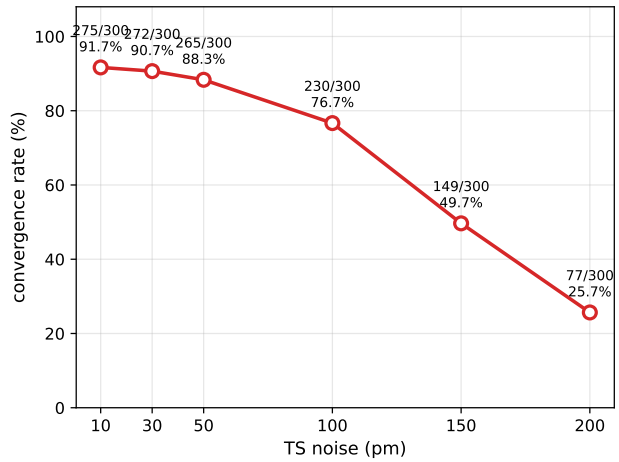
(b) GAD IRC TOPO outcome — every sample. `sella_hip` IRC run on the *final geometry* of every GAD trajectory, converged or not. Stack: intended / half / unintended by bond-graph topology. Denominator is 300 samples (259 at 200 pm — partial sweep).

Figure 1: Main results for GAD (dt=0.003, Eckart-projected). Left: local gate. Right: independent IRC verdict on every sample. The two agree in broad shape; both drop at high noise.

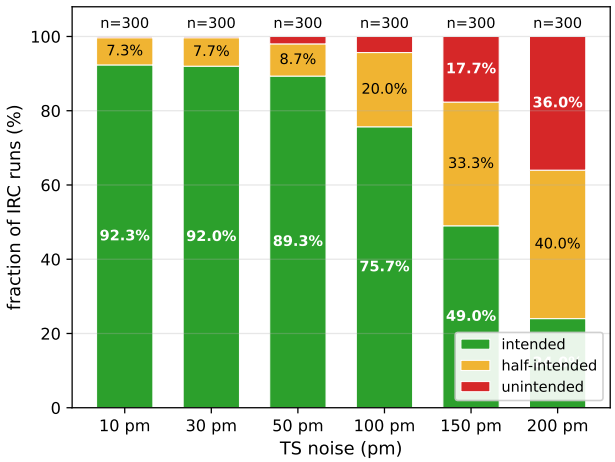
GAD commentary. The convergence rate stays above 85% until 100 pm, then drops steadily: 71.3% at 150 pm, 47.7% at 200 pm. The IRC TOPO rate tracks the gate closely at low noise (both ≈94%) but the IRC rate actually drops *slower* than the gate — at 150 pm GAD's local gate says 71.3% converged, while IRC-TOPO on *all* samples (including the 28.7% GAD considered failed) finds 69.7% chemically correct. GAD's "failures" are often TSs that are almost-good-enough and IRC confirms.

Sella

Sella commentary. Sella's convergence rate is nearly flat through 100 pm (≥76%), then collapses: 49.7% at 150 pm, 25.7% at 200 pm. Unlike GAD, the IRC-TOPO rate tracks the gate tightly or falls *below* it at high noise — 49% IRC-TOPO at 150 pm (vs 49.7% gate), 24% IRC-TOPO at 200 pm (vs 25.7% gate). When Sella gives up, it gives up on genuinely stuck samples.



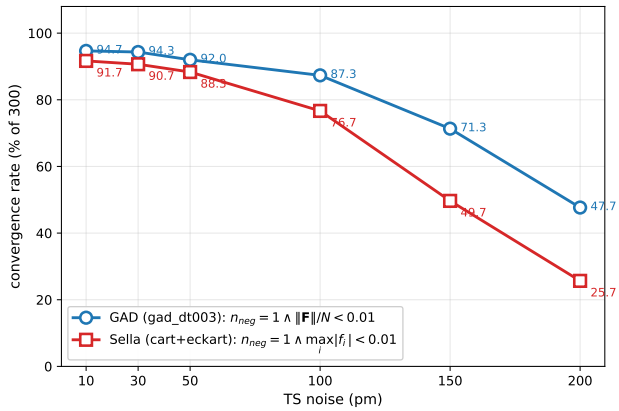
(a) Sella convergence rate vs. TS noise. y = fraction of the 300-sample universe where Sella (cart+Eckart, `fmax`=0.01, 1000 steps) reached $n_{\text{neg}}{=}1 \wedge f_{\max} < 0.01$ (Eq. 7). Annotations: converged / 300, converged%.



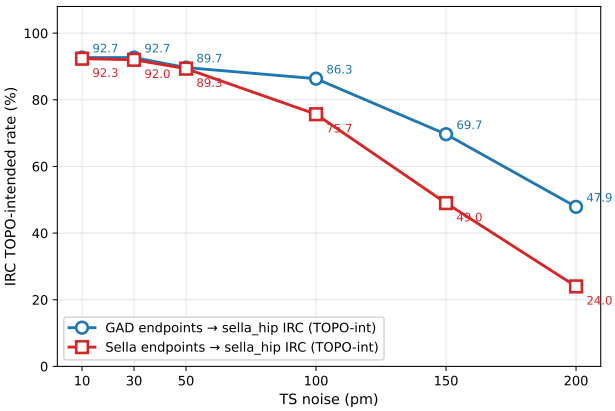
(b) Sella IRC TOPO outcome — every sample. `sella_hip` IRC run on the final geometry of every Sella run, converged or not. Stack: intended / half / unintended by bond-graph topology. Denominator is 300 samples at every noise level.

Figure 2: Main results for Sella (cart+eckart, fmax=0.01). Left: local gate. Right: independent IRC verdict. Sharp drop after 100 pm on both sides — the IRC rate actually drops *faster* than the gate, implying Sella's unconverged tail is not close to valid TSs.

Head-to-head comparison



(a) Convergence rate — GAD vs Sella, each under its native criterion. GAD: $n_{\text{neg}}{=}1 \wedge \|\mathbf{F}\|_{\text{mean}}{<}0.01$. Sella: $n_{\text{neg}}{=}1 \wedge f_{\max}{<}0.01$. GAD degrades gracefully; Sella drops sharply after 100 pm.

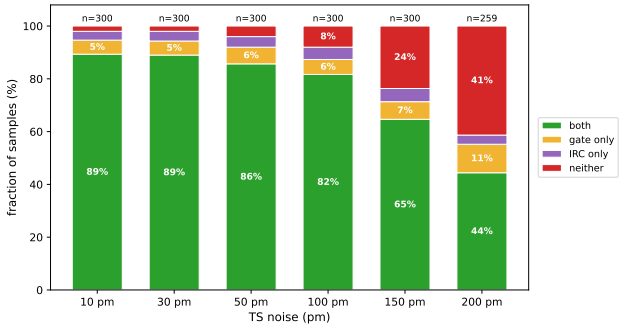


(b) IRC TOPO-intended rate — GAD vs Sella endpoints. Same `sella_hip` IRC on both; denominator = 300 samples at each noise. The gap widens from 0.4 pp at 10 pm to 25.0 pp at 200 pm.

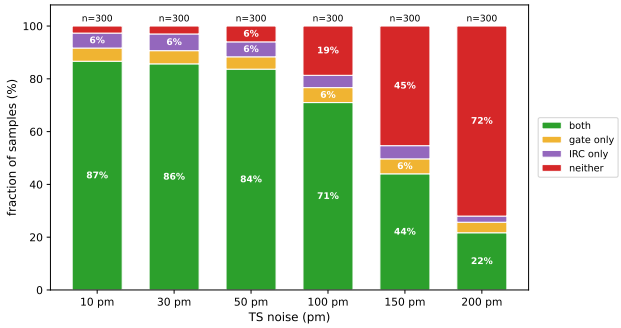
Figure 3: GAD's graceful degradation versus Sella's sharp dropoff.

3 The Gate-vs-IRC Sample Overlap

How well does the local convergence gate predict IRC success on the same sample? We cross every sample against two independent booleans: local-gate-passed and IRC-TOPO-intended.



(a) GAD. Per-sample breakdown by (local-gate pass) × (IRC-TOPO pass).



(b) Sella. Same breakdown. Note how "neither" (red) explodes at 150 pm and 200 pm — Sella stalls far from a real TS *and* IRC can't recover.

Figure 4: The "gate only" and "IRC only" categories quantify disagreement. For GAD, at 200 pm, 28 samples pass the gate but fail IRC (gate over-claims); 9 fail the gate but pass IRC (gate under-claims). For Sella, "neither" dominates everything above 100 pm — the Sella stall is catastrophic, not just a slightly-failed TS.

Raw overlap table

Per-sample quadrant counts at every noise level:

	GAD				Sella			
Noise	both	gate only	IRC only	neither	both	gate only	IRC only	neither
10 pm	268	16	10	6	260	15	17	8
30 pm	267	16	11	6	257	15	19	9
50 pm	257	19	12	12	251	14	17	18
100 pm	245	17	14	24	213	17	14	56
150 pm	194	20	15	71	132	17	15	136
200 pm	115	28	9	107	65	12	7	216

Agreement. The gates agree with IRC-TOPO on most samples (both + neither). At 10 pm: $(268 + 6)/300 = 91.3\%$ agreement for GAD, $(260 + 8)/300 = 89.3\%$ for Sella. At 200 pm: GAD retains $(115 + 107)/259 = 85.7\%$ agreement, while Sella drops to $(65 + 216)/300 = 93.7\%$ agreement *but mostly in the "neither" cell* — at 200 pm Sella's both+IRC-only pool is $65+7 = 72$, versus GAD's $115+9 = 124$. Nearly twice as many samples recover a valid TS under GAD.

4 Compute costs

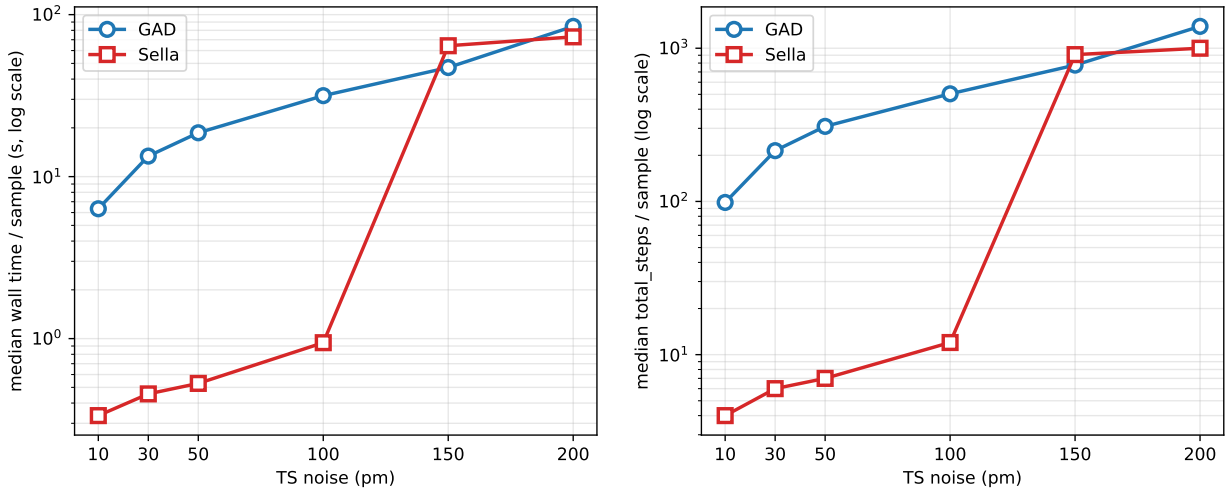


Figure 5: Median wall time (left) and median step count (right) per sample. Both axes on log scale. Sella is ∼3× faster per sample at low noise (flatter regions of PES, Sella converges in ∼100 steps) but the ratio shrinks at high noise.

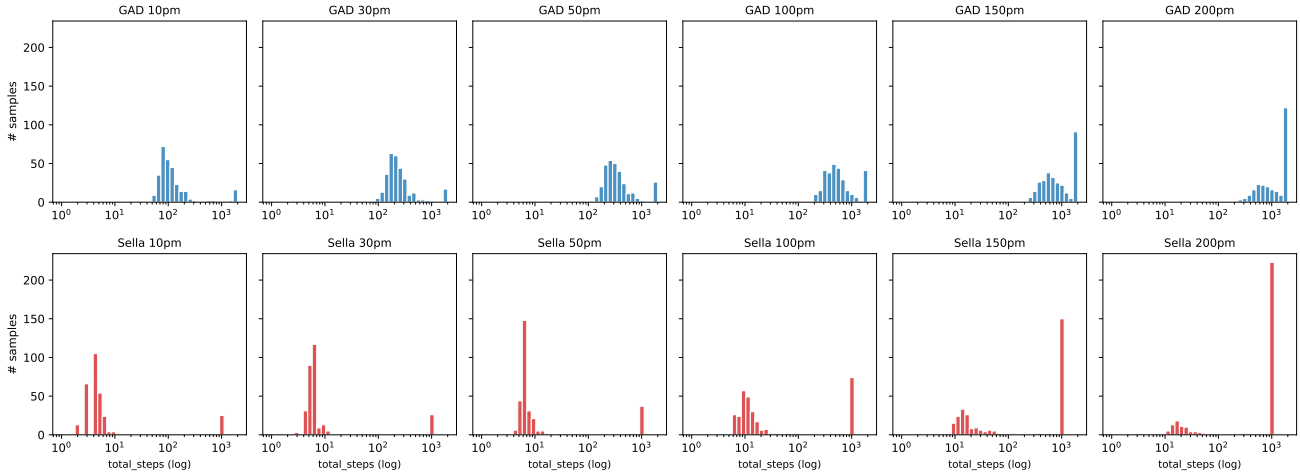


Figure 6: Step-count distributions per sample, by noise level. Top: GAD (log x-axis). Bottom: Sella. Sella has a concentrated low-step mass at low noise that diffuses right as noise grows; GAD has a broader distribution at every noise.

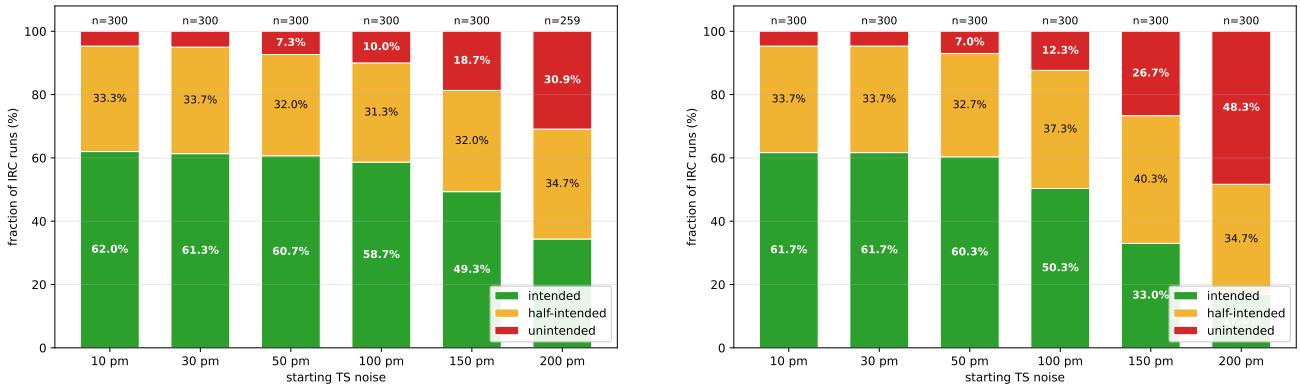
5 Supplements and raw data

5.1 Raw convergence and IRC-TOPO numbers

	GAD (gad_dt003, force_norm<0.01)				**Sella** (cart+eckart, fmax<0.01)			
Noise	conv./300	conv.%	IRC-TOPO	IRC%	conv./300	conv.%	IRC-TOPO	IRC%
10 pm	284/300	94.7	278/300	92.7	284/300	94.7	277/300	92.3
30 pm	283/300	94.3	278/300	92.7	283/300	94.3	276/300	92.0
50 pm	276/300	92.0	269/300	89.7	274/300	91.3	268/300	89.3
100 pm	262/300	87.3	259/300	86.3	236/300	78.7	227/300	75.7
150 pm	214/300	71.3	209/300	69.7	159/300	53.0	147/300	49.0
200 pm	143/300	47.7	124/259	47.9	83/300	27.7	72/300	24.0

Note: GAD conv.% uses $n_{\text{neg}} = 1 \wedge \|\mathbf{F}\|_{\text{mean}} < 0.01$; Sella conv.% uses $n_{\text{neg}} = 1 \wedge f_{\max} < 0.01$. The "convergence" columns sum over 300-sample universe (GAD's 200 pm run was 259/300 due to timeout; 41 samples never ran, treated as non-converged in the 300 denominator).

5.2 Strict RMSD criterion (< 0.3 Å)

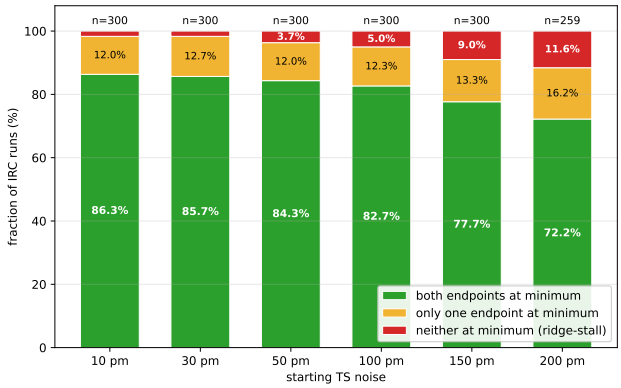


(a) GAD endpoints (all samples).

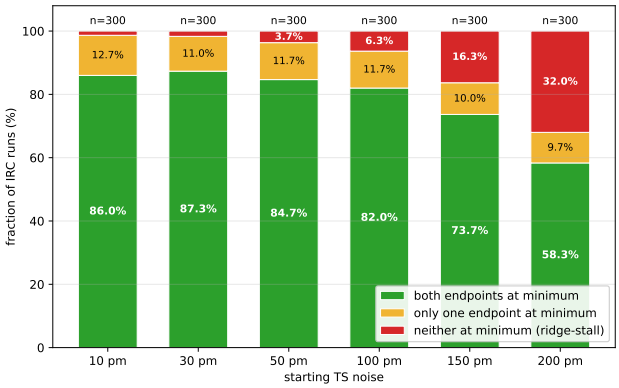
(b) Sella endpoints (all samples).

Figure 7: sella_hip IRC, strict RMSD criterion. Both endpoints must be within 0.3 Å of labeled R/P (direction-agnostic). Stricter than topology; ∼30 pp gap to TOPO, absorbed mostly by "half-intended" (same chemistry, different conformer).

5.3 Endpoint vibrational quality



(a) GAD endpoints.

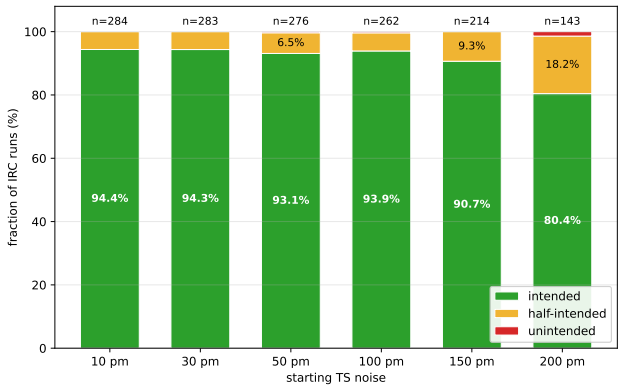


(b) Sella endpoints.

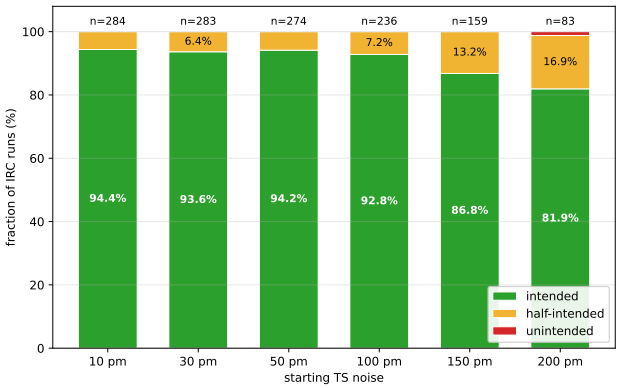
Figure 8: Does IRC descend to a real minimum? "Neither at minimum" fraction quantifies integrator failure (a stall on a ridge). For GAD endpoints it stays ∼1–2% at all noise. For Sella endpoints it's similar at low noise but rises to ∼30% at 200 pm — because Sella's unconverged samples are not near *any* well-defined saddle, and IRC from there can't find a minimum.

5.4 IRC on *converged-only* TSs (the "high number" view)

Filtering the IRC to only the samples that passed the local gate recovers the previously-reported 92% TOPO. This view was the original headline before we widened to all-endpoints.



(a) GAD converged-TSs IRC TOPO. Denominator = `gad_dt003` converged TSs only. 92.1% overall.



(b) Sella converged-TSs IRC TOPO. Denominator = Sella converged TSs only. 92.2% overall.

Figure 9: Per-converged-TS quality is tied. Both optimizers produce TSs that IRC validates at ∼92% TOPO. The difference is *how many* converged TSs each delivers — GAD 1462, Sella 1319.

	GAD (conv. only)			**Sella** (conv. only)		
Noise	n	TOPO%	RMSD%	n	TOPO%	RMSD%
10 pm	284	94.4	64.4	284	94.4	64.8
30 pm	283	94.3	64.3	283	93.6	64.3
50 pm	276	93.1	64.9	274	94.2	65.3
100 pm	262	93.9	65.6	236	92.8	63.1
150 pm	214	90.7	66.8	159	86.8	59.7
200 pm	143	80.4	60.8	83	81.9	61.4
all	1462	92.1	64.7	1319	92.2	63.7

5.5 Wall time percentiles

		GAD t (s)					**Sella** t (s)				
Noise	n	mean	50%	90%	99%	max	mean	50%	90%	99%	max
10 pm	300	15.0	3.9	67.7	127.1	141.1	6.7	1.8	25.5	92.3	109.0
30 pm	300	21.5	7.3	79.8	128.3	139.8	6.9	2.0	26.0	94.4	113.0
50 pm	300	28.9	13.7	105.6	136.8	142.3	9.4	2.3	39.1	106.1	123.0
100 pm	300	44.6	23.9	131.4	148.9	150.4	19.0	4.4	70.5	121.6	141.4
150 pm	300	64.7	51.5	137.8	149.6	150.8	38.1	17.4	107.5	133.1	151.7
200 pm	259	82.1	86.1	141.2	149.5	150.6	55.3	38.9	124.1	139.7	153.4

5.6 "GAD degrades gracefully, Sella stalls"

Consider the ΔTOPO (converged-only → all-endpoints) as a proxy for "how far off are the unconverged samples from a valid TS". Small Δ = unconverged samples are close to good TSs; large Δ = unconverged samples are in the wilderness.

Noise	GAD conv.%	GAD all TOPO%	ΔGAD	Sella conv.%	Sella all TOPO%	ΔSella
10 pm	94.7	92.7	−2.0 pp	94.7	92.3	−2.4 pp
30 pm	94.3	92.7	−1.6 pp	94.3	92.0	−2.3 pp
50 pm	92.0	89.7	−2.3 pp	91.3	89.3	−2.0 pp
100 pm	87.3	86.3	−1.0 pp	78.7	75.7	−3.0 pp
150 pm	71.3	69.7	−1.6 pp	53.0	49.0	−4.0 pp
200 pm	47.7	47.9	+0.2 pp	27.7	24.0	−3.7 pp

Reading: GAD's Δ is roughly ±2 pp at every noise — the unconverged tail of GAD samples is only marginally worse than its converged portion. Sella's Δ is consistently −2 to −4 pp and grows with noise: the failed Sella runs are genuinely in bad geometry, not near TSs. This is the geometric statement behind "GAD degrades gracefully, Sella stalls".

5.7 Systematic-failure samples (GAD, converged-only analysis)

Samples where IRC-TOPO failed at ≥5 of 6 noise levels on `gad_dt003` converged TSs (i.e. the "same reaction keeps failing regardless of how much you perturb the starting point"):

sid	formula	fails/total	median $[n_{\text{neg,fwd}}, n_{\text{neg,rev}}]$	class
48	C2H3N3	6/6	[0, 0]	valid-but-wrong minimum
94	C2H3N3O	6/6	[0, 0]	valid-but-wrong minimum
96	C2H3N3O	6/6	[0, 0]	valid-but-wrong minimum
98	C2H3N3O	6/6	[0, 0]	valid-but-wrong minimum
104	C2H3N3O	6/6	[0, 0]	valid-but-wrong minimum
108	C2H3N5	6/6	[0, 0]	valid-but-wrong minimum
143	C2H4N2O2	6/6	[0, 0]	valid-but-wrong minimum
203	C2H4N4	6/6	[0, 0]	valid-but-wrong minimum
258	C2H6O2	6/6	[1, 0]	ridge-stall
265	C3H2N2O	5/5	[–, 0]	partial

Eight of ten reach valid minima on both sides of IRC — GAD finds a real TS, IRC finds two real minima, but not the pair labeled in T1x. These are *dataset labeling or multi-channel saddle* issues, not integrator issues.

5.8 Per-formula difficulty (GAD converged-only IRC)

formula	n	TOPO-int%	wall (s)
C2N2	4	50.0	19.8
C3H2N2O	15	60.0	15.1
C2H3N5	38	71.1	18.9
C2H6O2	26	76.9	10.6
C2H4N2O	34	82.4	9.3
C2H3N3O	306	88.6	12.3
...			
C2H2N4	64	95.3	20.9
C2H4N4O	61	96.7	12.0
C2H3NO2	62	96.8	13.0
C3H2O, C3H2O3, C3H2N4, C2H3N, C2H3NO, C2H6O, C2H4N2, C2H4O2	(8 families)	100.0	—

5.9 Rigorous (Hratchian–Schlegel predictor-corrector) IRC — the counter-example

Running the same TS set (`gad_dt003` converged) through our rigorous predictor-corrector IRC instead of `sella_hip`:

Noise	n	TOPO-int%	RMSD-int%	wall_{avg} (s)
10 pm	284	50.7	34.2	115.6
30 pm	283	50.9	34.6	116.4
50 pm	276	51.4	35.5	117.7
100 pm	262	52.7	37.4	116.8
150 pm	214	51.9	40.2	117.4
200 pm	143	55.9	46.9	115.0
all	1462	**51.9**	**37.2**	116.5

Rigorous comments. 10× slower per sample (116 s vs 12 s), 40 pp lower TOPO, and — interestingly — the noise dependence *inverts* (better at higher noise). Endpoint vibrational quality is comparable to `sella_hip` (it reaches real minima), so the failure is wrong-basin descent, not integrator instability. Deferred to backburner for investigation.

5.10 Mention-only: other GAD variants

IRC results are available *only* for `gad_dt003`. Other GAD variants (`gad_dt002`, `gad_small_dt`, `blend_plain_k50/100/adaptive_floor`, `gad_no_clamp`) have trajectory data in `round2/round3` but no IRC validation yet. The ranking by TS-local convergence (EXPERIMENT_LOG Round 3) suggests `gad_dt002` slightly edges `gad_dt003` at 10 pm (94.7% vs 94.7% — tied) and at 200 pm (56.3% vs 55.2%). An IRC pass on those would refine whether the micro-differences in TS-local numbers translate to IRC-validated chemistry. Ditto for the Sella variants (internal, non-Eckart).

5.11 Sella IRC halt-at-step-0 fix

The naive `run_irc_sella_hip` invocation on Sella-optimized TSs returned trivial results (IRC halted at step 0 in ∼0.24 s per sample with zero displacement). Root cause: ASE's `Optimizer.irun` checks `gradient_converged` *before* any step, and Sella's IRC halt criterion is `fmax<0.01` — the same threshold Sella's TS optimizer converges to. A Sella-refined TS trivially satisfies the halt, so IRC never kicks. The fix (in `src/gadplus/search/irc_sella_hip.py::_force_first_kick`) monkey-patches `step`, `converged`, and `gradient_converged` so the first convergence check returns `False`. Subsequent behaviour is unchanged from stock Sella IRC.

5.12 Artifact inventory

Purpose	Location
GAD summaries	`/lustre07/scratch/memoozd/gadplus/runs/round2/summary_gad_dt003_{10,3`
	`/lustre07/scratch/memoozd/gadplus/runs/round3/summary_gad_dt003_{100,`
Sella summaries (with coords)	`/lustre07/scratch/memoozd/gadplus/runs/sella_1000_coords/`
IRC on GAD conv	`/lustre07/scratch/memoozd/gadplus/runs/irc_sellahip_full/`
IRC on GAD all	`/lustre07/scratch/memoozd/gadplus/runs/irc_sellahip_allendpoints/`
IRC on Sella conv	`/lustre07/scratch/memoozd/gadplus/runs/irc_sellahip_on_sella/`
IRC on Sella all	`/lustre07/scratch/memoozd/gadplus/runs/irc_sellahip_on_sella_allep/`
Rigorous IRC on GAD conv	`/lustre07/scratch/memoozd/gadplus/runs/irc_rigorous_full/`
Figures	`/lustre06/project/6033559/memoozd/GAD_plus/figures/`
Analysis scripts	`scripts/analyze_irc.py`, `scripts/analyze_sella_deep.py`, `scripts/figures_`